

EVALUATING THE EFFECTIVENESS OF AI-GENERATED DEVICE REPRESENTATIONS IN DIGITAL TWIN VISUALIZATIONS

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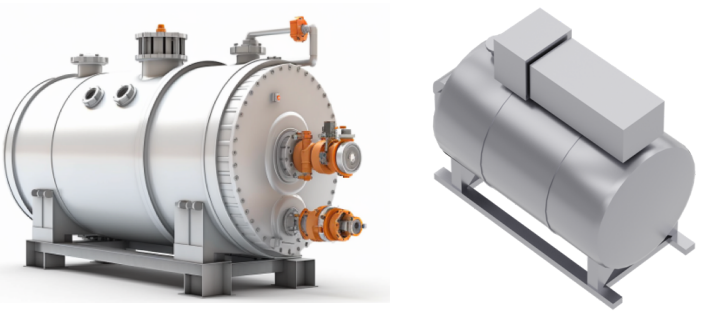
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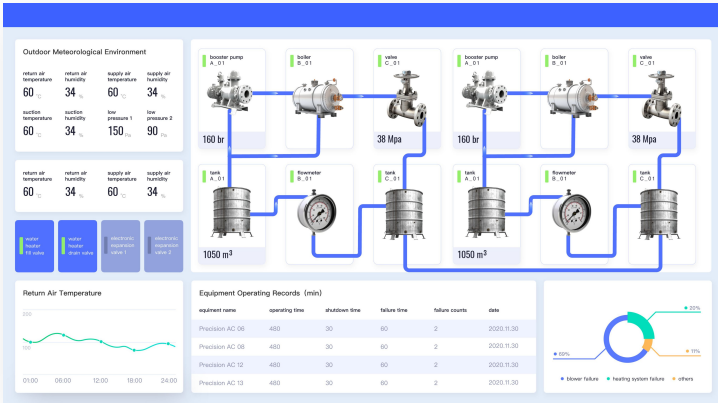
HIGHLIGHTS

AI-generated device is better than 3D modelling

Feedback from 31 professional users reveals that AI-generated device representations outperform traditional 3D modeling without affecting user information perception.



For individual device representation, AI-generated **outperforms** 3D modelling in terms of **aesthetics, clarity, realism, and accuracy**.

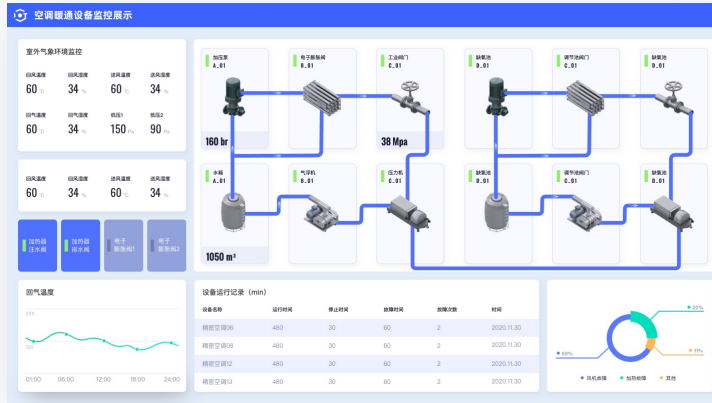


80.65 % of users reported that the AI-generated device representation did not affect their data visualization reading

BACKGROUND

A digital twin is a virtual model of a physical entity or process, designed to gather real-world data for accurately representing, validating, and simulating current and future performance. Device representations are key visualization elements for users to perceive digital twins. But they require considerable design effort and time-consuming drawing processes. We hope to improve efficiency of these processes with AI-generated device representations alternating designer-drawing ones. However, this alternative work requires examining whether AI-generated device representations will affect user perception. Currently most validations of AIGC contents are based on cross-database comparison methods or keyword matching, but more research for human perception is not conducted.

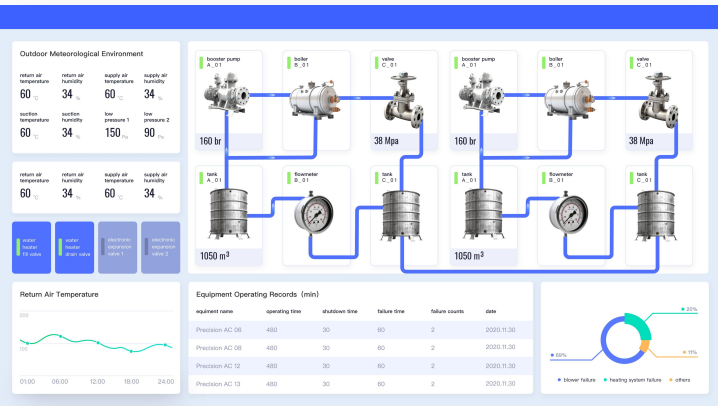
↓ Typical digital twin Visualization with device representation



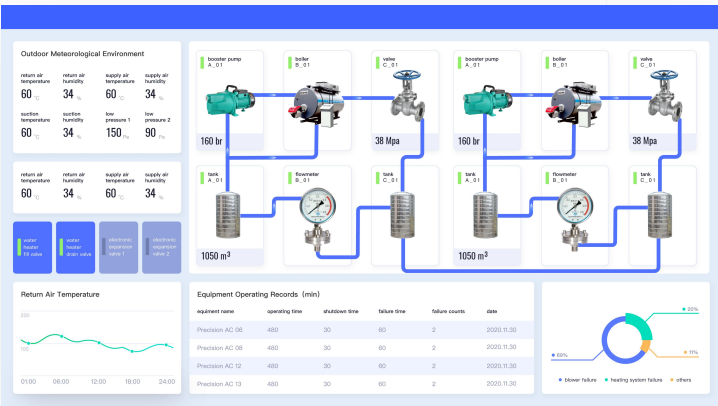
METHODOLOGY

Methods of device representation

AI-Generated



Photography



3D modeling



Evaluation method

Accuracy

Correctness of identifying device representations in 3 styles

Clarity, realism, aesthetics

Comparing the 3 styles and chose the one that best met expectations

Display on dashboards

Identify the AI-generated one from 3 dashboards. And judge if it would affect data perceiving.

Target User Group

We aimed emphasize AIGC's potential beyond generic domains. Therefore, we chose expert users in the digital twin related fields.

Designers

Developers

Operators

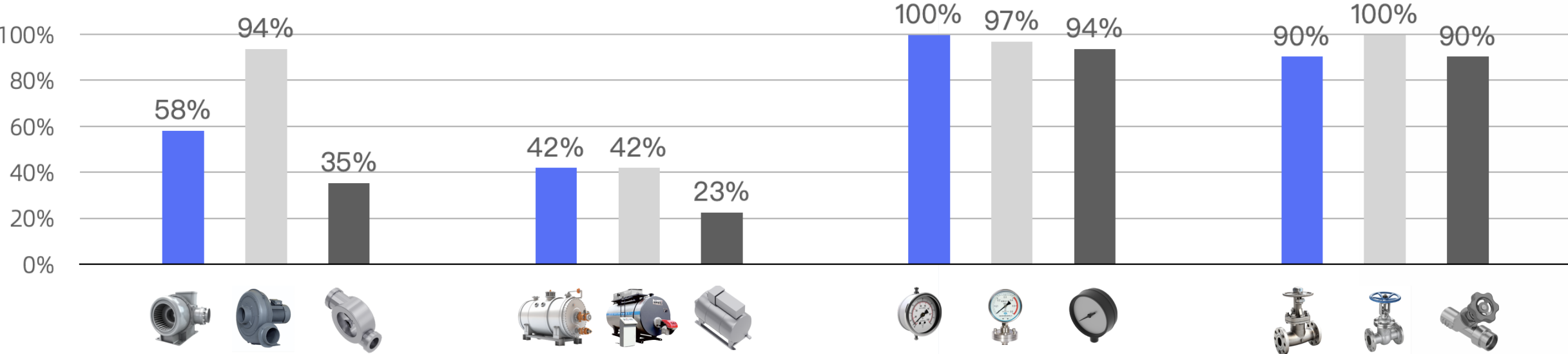
Architects

EXPERIMENT RESULT

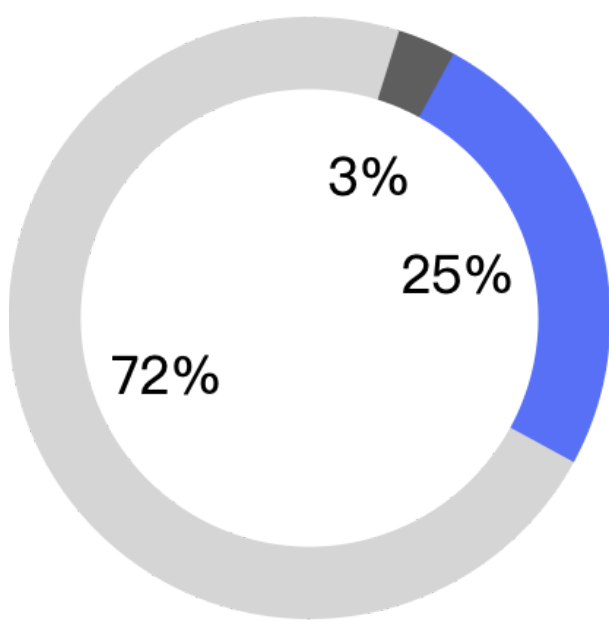
Individual device representation

■ AI-Generated ■ Photography ■ 3D modeling

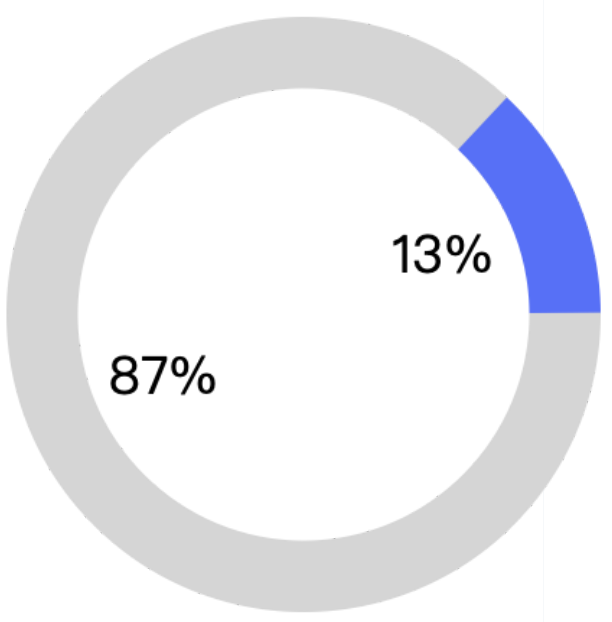
Accuracy



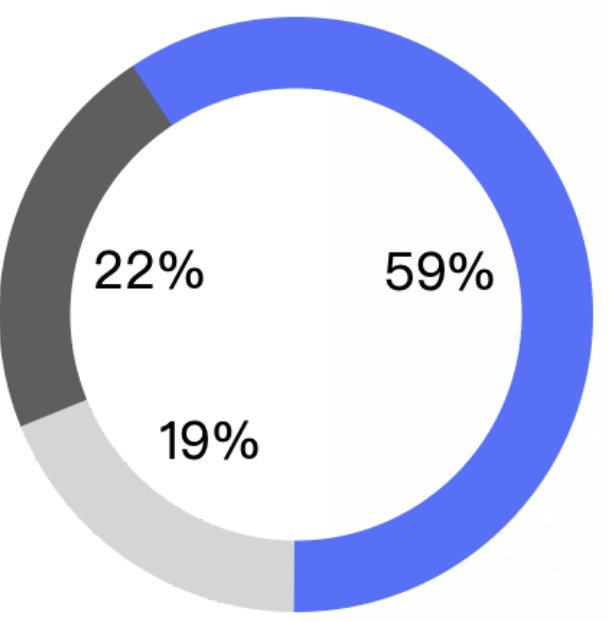
Clarity



Realistic

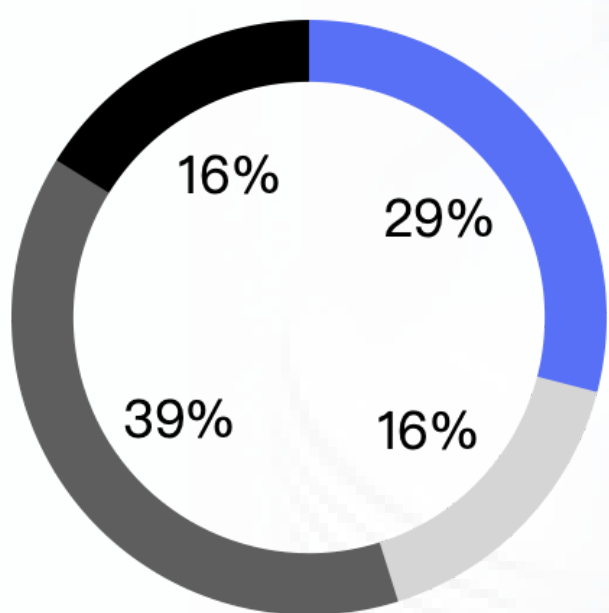


Aesthetic

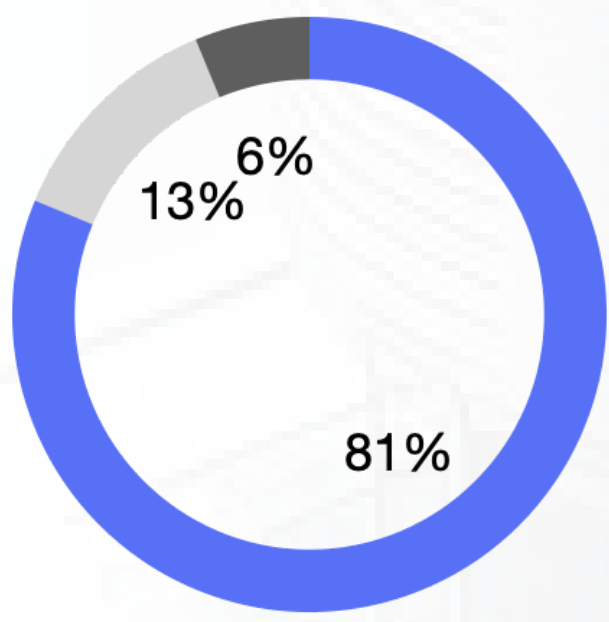


Integration with graphical user interface

■ AI-Generated ■ Photography ■ 3D modeling



29.03 % of users correctly identified which graphical dashboard they thought was AIGC-generated,



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DISCUSSION

Device representation is a relatively concrete concept

User perception goes through from the physical world to the visualization screen. The diffusion model-based AIGC tool tends to generate many realistic details. In contrast, 3D modeling intentionally removes details to fit simplism design, leading to some information loss. Therefore, AI-generated device representation has a natural advantage in digital twin visualization scenarios. This also explained why real photos outperform in terms of accuracy and clarity. As one of the interviewees (a solution architect) pointed out, "For workers, abstract models would be considered too different from reality. Industrial equipment has distinct features. If the device's characteristics are not prominent, it is not representative for users."

Information is conveyed in multi-modals

Device representation often appears alongside device names and data. visual elements on graphical interfaces such as line graphs and texts help attenuate the randomness and incorrectness details of AI-generated devices. As another interviewee (a product manager) said: "Use actual device images with text descriptions with better fit the comprehension of actual users."

Static representation is enough in most cases

In 2D data visualization, device representation does not require exploded views or animations. Thus, the advantages of 3D modeling are not demonstrated. In manufacture contexts, there are more 2D display than those requiring 3D animations.

CONCLUSION

Conclusion

In this paper, we compared and analyzed the impact of device representations created using different methods in the field of digital twin visualization. The results showed that AI-generated device representations were superior to 3D modeling in terms of accuracy, clarity, and aesthetics, and did not affect users' information acceptance on the interface.

Future work

This experiment was based on questionnaire. Future research is suggested to be complete in application environments with quantitatively ergonomic measurements in the effectiveness of visualization. Meanwhile, this study only compared 2D graphical visualization, and further experiments and comparative analyses need to be conducted for 3D spatial digital twins.

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